A PRACTITIONER'S FIELD GUIDE · BEGINNER TO INDUSTRY EXPERT

Tokenization & Digital Currency

How money and assets move onto programmable ledgers — stablecoins, CBDCs, tokenized deposits, tokenized funds and bonds, and the market infrastructure behind them. From first principles to the institutional frontier.

Digital Assets & Tokenization Masterclass
Edition 1.0 · 2026.10
Companion to the Treasury cinematic course

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01 · EXECUTIVE SUMMARY

How to Use This Guide & Executive Summary

The plumbing of finance is being rebuilt on programmable ledgers. Money is becoming a token — stablecoins, tokenized deposits and central-bank digital currencies — and so are the assets money buys, from Treasury bills to private credit. This guide takes you from "what is a token" to designing an institutional tokenization and on-chain funding operating model.

Who this is for

Anyone who wants to understand — and speak credibly about — tokenization and digital money: bankers, treasurers and operations staff whose rails are changing under them; technologists and product managers building digital-asset platforms; investors, policymakers and journalists trying to separate substance from hype. No crypto background is assumed, and no prior finance degree. Every term is defined the first time it appears, and the focus throughout is the *institutional, regulated* use of this technology, not speculation.

How the guide is built

The twelve chapters climb a ladder. Chapters 2–3 are the foundations: what a token and a ledger actually are, and a clean map of every form of digital money. Chapters 4–6 cover digital money itself: stablecoins, central-bank digital currencies, and tokenized bank deposits and money-market funds. Chapters 7–9 cover tokenized assets and markets: real-world-asset tokenization, the settlement and smart-contract infrastructure, and what all of this does to the treasury and funding function. Chapters 10–11 cover risk and mastery: the real risk-and-control map and regulation, then what a best-in-class institutional operating model looks like. Chapter 12 is a glossary and reading list.

HOW TO READ IT

Read linearly the first time — each chapter builds on the last. Every mechanism is shown with a named real-world example and, where there are economics, a fully worked calculation. Every new term sits in a definition box, and every chapter ends with a "Best in class" box describing what leading institutions actually do. Afterwards it works as a reference.

The one-paragraph version

To "tokenize" is to represent a claim — a dollar, a bond, a fund share — as a programmable entry on a shared ledger, so it can move and settle under rules written in software. Do that to money and you get stablecoins (private tokens backed by reserves), tokenized deposits (bank money on a ledger, redeemable at par) and CBDCs (central-bank money in digital form). Do it to assets and you get tokenized Treasuries, bonds and funds that settle in seconds and pay coupons by smart contract. The prize is atomic settlement (both legs of a trade move together or not at all), collateral that moves in minutes, 24/7 markets and automated servicing — which

together compress settlement risk, trapped liquidity and operational cost. The constraints are equally real: smart-contract and custody risk, the quality of the settlement asset, legal finality, and prudential capital. The institutions winning are the ones treating this as a plumbing upgrade settled in regulated money, not a speculative asset class.

What you will be able to do by the end

- Place any instrument correctly on the digital-money map and say who bears the risk.
- Explain how a stablecoin holds its peg, work the arbitrage, and read a reserve report.
- Distinguish a stablecoin, a tokenized deposit and a CBDC — and why the difference matters.
- Walk the lifecycle of a tokenized bond or fund and quantify the issuance and settlement savings.
- Explain atomic DvP and compute the settlement risk and capital it removes.
- Show how tokenized collateral and intraday repo change the treasury and funding desk, with numbers.
- Map the real risks and the capital and regulatory treatment (Basel, MiCA, the GENIUS Act).

QUESTIONS THIS GUIDE ANSWERS

What does “tokenization” actually change, beneath the jargon? What really backs a stablecoin, and what makes one break? How is a CBDC different from the money already in your bank app? Why are the largest asset managers and banks putting money-market funds and deposits on-chain? And what separates an institution merely running pilots from one with a genuine, scaled, best-in-class digital-asset operating model?

02 · FOUNDATIONS

Money, Ledgers & What Tokenization Really Means

Before you can reason about stablecoins, central-bank digital currencies or tokenized deposits, you need a precise mental model of three things: what a ledger is, what a token actually is, and what the verb "tokenize" really does. This chapter builds that model from first principles and then proves it with arithmetic you can follow line by line.

A token is a claim recorded on a programmable ledger

Start with the humble ledger. A ledger is simply a record of who owns what and who owes whom. A bank's core banking system is a ledger; your brokerage statement is a ledger; the deed registry at a county courthouse is a ledger. For most of financial history these ledgers have been centralized: one trusted operator holds the authoritative copy and everyone else trusts that copy.

> Token — a transferable entry on a ledger that represents a claim: a claim on an asset, on a quantity of money, or on a bundle of rights. The token is not the asset itself; it is the ledger's authoritative record that a particular holder controls that claim.

The word token frequently misleads newcomers, who imagine a coin-like object moving around a network. Nothing moves. A token transfer is an update to balances in a shared ledger: one account's number goes down, another's goes up, and the ledger's rules guarantee the two always reconcile. What makes modern tokens interesting is that the ledger is programmable — the rules governing how entries can change are themselves code that runs automatically.

DLT versus a blockchain

> Distributed ledger technology (DLT) — any system in which multiple parties each hold and maintain synchronized copies of the same ledger, with an agreed protocol (a consensus mechanism) for keeping those copies identical. There is no single master copy.

> Blockchain — a specific kind of DLT in which updates are grouped into batches ("blocks") that are cryptographically chained to the previous batch, so that altering an old record would break every subsequent link. Every blockchain is a DLT, but not every DLT is a blockchain.

The distinction matters because much of institutional tokenization runs on DLT that is deliberately not a public blockchain. A consortium of banks might run a shared ledger among themselves without ever grouping

transactions into a public chain of blocks. When a vendor says "blockchain," ask whether they mean a specific chained-block design or simply a shared, synchronized ledger.

Permissioned versus permissionless

> Permissionless ledger — anyone may read the ledger, submit transactions, and participate in validating them without approval. Bitcoin and Ethereum are permissionless.

> Permissioned ledger — only approved, identified participants may validate transactions (and sometimes only approved parties may even read or transact). Membership is controlled by an operator or consortium.

Retail cryptocurrencies live on permissionless ledgers; most regulated financial institutions, bound by know-your-customer and anti-money-laundering rules, favor permissioned ledgers where every counterparty is identified. The spectrum is not binary — some designs are permissioned for validation but permissionless for reading, and others sit in between.

On-chain versus off-chain

> On-chain — data or logic that lives on, and is enforced by, the ledger itself. An on-chain transfer is final the moment the ledger's rules accept it.

> Off-chain — data, assets or legal rights that live in the traditional world (a bank account, a vault, a court-enforceable contract) and are merely *referenced* by the ledger. The ledger records a claim; the enforcement happens elsewhere.

This is the single most important seam in all of tokenization. A token for US dollars is only as good as the off-chain promise that someone will hand you a real dollar when you redeem it. The ledger can be flawless and the token can still be worthless if the off-chain backing fails — a theme we return to repeatedly when we dissect stablecoin depegs in Chapter 04.

What "tokenize" actually does

To tokenize an asset or a claim is to represent it as a transferable ledger entry governed by programmable rules. Tokenization does three concrete things at once:

1. Represents — it creates an authoritative digital record of ownership or entitlement.
2. Mobilizes — it makes that record transferable, often instantly and continuously, without a central transfer agent re-keying the change.

3. Programs — it attaches rules (transfer restrictions, automatic interest accrual, atomic delivery-versus-payment) that execute without manual intervention.

> Atomic settlement — a transaction in which all legs either complete together or none complete at all. Delivery of the asset and payment for it happen as one indivisible event, eliminating the risk that one party pays and the other fails to deliver.

Native tokens versus tokenized claims

> Native token — a token whose value and existence are intrinsic to its own ledger; it is not a claim on anything off-chain. Bitcoin and ether are native tokens — holding one is holding the thing itself, not a receipt for something external.

> Tokenized claim — a token that is a digital wrapper around an off-chain asset or liability (a dollar deposit, a Treasury bill, a share). Its value depends entirely on the enforceability of the underlying claim.

Stablecoins, tokenized deposits and tokenized funds are all tokenized claims. This is why their risk analysis always comes back to the quality of the backing and the credibility of redemption, never to the elegance of the code alone.

Cryptocurrency, security token, and digital money

Three categories newcomers routinely conflate:

- Cryptocurrency — typically a native token on a permissionless ledger (e.g. ether), valued by supply and demand, not a claim on an issuer.
- Security token — a tokenized claim representing a regulated financial instrument (equity, debt, a fund share) and subject to securities law. A tokenized money-market fund (Chapter 06) is a security token.
- Digital money — a tokenized claim intended to be used *as money* — stable in value and redeemable at par. Stablecoins, CBDCs and tokenized deposits are digital money (Chapters 03–06).

WORKED EXAMPLE

A ledger-transfer walkthrough. Alice holds 1,000 tokenized dollars; Bob holds 0. Alice sends Bob 250.

```
STATE BEFORE          balance (tokens)
  Alice ............... 1,000
  Bob   ...............     0
  TOTAL ............... 1,000

TRANSACTION: Alice → Bob, 250
  Rule check: Alice balance (1,000) ⩾ amount (250)?  YES
  Debit Alice: 1,000 - 250 = 750
  Credit Bob:      0 + 250 = 250

STATE AFTER           balance (tokens)
  Alice ...............   750
  Bob   ...............   250
  TOTAL ............... 1,000   (conserved)
```

The total is conserved (750 + 250 = 1,000) — no tokens were created or destroyed, only reassigned. Once the ledger's consensus rule accepts the block containing this update, the transfer is final: there is no clearing lag and no reversal. Contrast this with a card payment, which is "authorized" instantly but can be charged back for months.

WORKED EXAMPLE

Fractionalization and reach. A private-credit note has a face value of $1,000,000 and a traditional minimum investment ticket of $1,000, so it can be split among at most 1,000,000 / 1,000 = 1,000 holders. Tokenize it into units of $50 each: now the same $1,000,000 supports 1,000,000 / 50 = 20,000 holders. The investor base reachable per million dollars of face value widens by a factor of 1,000 / 50 = 20×. If a distributor can realistically reach, say, 400,000 retail investors who will each commit $50, the tokenized structure can absorb 400,000 × $50 = $20,000,000 of demand that the $1,000-minimum structure could never have tapped at the small end.

BEST IN CLASS

The institutions that tokenize well treat the off-chain seam as the whole game. JPMorgan's permissioned ledger (Kinexys, Chapter 06) pairs every on-chain deposit token with an identified, regulated bank deposit and legally enforceable redemption, so the token inherits the bank's balance-sheet credibility rather than relying on code alone. The lesson for a beginner: judge any token first by the strength and enforceability of its off-chain claim, and only second by its technology.

03 · THE DIGITAL MONEY MAP

Stablecoins, CBDCs, Tokenized Deposits & e-Money

"Digital money" is not one thing — it is a family of instruments issued by different parties, backed in different ways, and carrying very different risks. This chapter gives you a single map: who issues each form, what stands behind it, whether it redeems at par, and where the settlement and credit risk actually sits.

Public money versus private money

> Public money — a direct liability of the central bank. It carries no credit risk because the issuer is the sovereign monetary authority. Physical cash, commercial banks' reserve balances at the central bank, and a central-bank digital currency are all public money.

> Private money — a liability of a private institution — a commercial bank deposit, an e-money balance, a stablecoin, or a tokenized deposit. It is a *promise* by that institution to pay you public money, and it is only as safe as the institution and its backing.

Almost all the money you use day to day is private money. The balance in your checking account is not cash in a vault; it is a claim on your bank. The genius of a well-functioning monetary system is that you never notice the difference — a theme captured by the singleness of money.

The singleness of money and par settlement

> Singleness of money — the principle that all the different forms of money in an economy trade at a fixed 1:1 rate with one another, so a dollar is a dollar regardless of whether it is cash, a deposit at Bank A, or a deposit at Bank B. Money is "singular" when you never have to ask *whose* dollar you are holding.

> Par settlement — settling a transfer at face value, one unit for one unit, with no discount. $1 of Bank A deposits must convert into $1 of Bank B deposits, not $0.99.

Singleness is not automatic — it is engineered. It rests on deposit insurance, central-bank backstops, supervision, and the ability to settle interbank obligations in public money (reserves). When that engineering fails, money loses its singleness: this is exactly what happened when USDC traded at ~$0.88 in March 2023

(Chapter 04). For a few days, one "dollar" of USDC was demonstrably not worth one dollar of bank deposits. A core policy worry about fragmented private digital money is that it reintroduces the pre-central-bank world where different issuers' notes traded at different prices.

The money-flower framework

The Bank for International Settlements popularized a four-property taxonomy — nicknamed the "money flower" — for classifying any form of money:

- Issuer — central bank or private institution?
- Form — a physical object or a digital/account entry?
- Accessibility — available to the general public, or restricted to banks (wholesale)?
- Technology — account-based (conventional database) or token-based (DLT)?

Every instrument below is just a particular combination of those four properties. Cash is central-bank-issued, physical, widely accessible, token-like. A retail CBDC is central-bank-issued, digital, widely accessible, and token-or-account-based. A stablecoin is privately issued, digital, widely accessible, and token-based.

The instruments, side by side

Instrument	Issuer	Backing	Par redemption?	Who can hold	Settlement asset
Central-bank reserves	Central bank	Sovereign (none needed)	n/a — is public money	Banks only (wholesale)	Itself
Physical cash	Central bank	Sovereign	n/a — is public money	Everyone	Itself
Retail CBDC	Central bank	Sovereign	Yes, by definition	Everyone (limits possible)	Itself (public money)
Bank deposit	Commercial bank	Bank balance sheet + insurance	Yes (insured up to limit)	Bank customers	Reserves (public money)
Tokenized deposit	Commercial bank	Same deposit, on a ledger	Yes, 1:1	Bank customers	Reserves (public money)
Stablecoin	Private issuer	Reserves (T-bills, cash, repo)	Issuer promise only	Token holders	Itself / off-chain reserves
E-money	Licensed e-money firm	Safeguarded client funds	Yes (safeguarded)	Account holders	Bank deposits

> E-money (electronic money) — a prepaid digital value balance issued by a licensed non-bank (think a prepaid-card or payments-app balance), legally required to be backed one-for-one by safeguarded client funds held at a bank. It predates stablecoins and is their closest regulated cousin.

Where each instrument sits

Read the table top to bottom and the risk gradient is clear. The top rows are public money — no credit risk. The middle rows (deposits, tokenized deposits, e-money) are private money inside the regulated perimeter, with insurance or safeguarding softening the credit risk. The bottom row — stablecoins — is private money whose safety depends entirely on reserves the issuer chooses and a redemption promise the issuer makes. The further down the table, the more the burden of proof shifts from "the sovereign stands behind it" to "trust this private balance sheet."

WORKED EXAMPLE

How $1 moves and settles under three forms. Alice pays Bob $1.

```
(A) BANK DEPOSIT (different banks)
  - Alice's Bank A debits her deposit  -$1
  - Banks settle via CENTRAL BANK RESERVES: A's reserves -$1, B's +$1
  - Bob's Bank B credits his deposit   +$1
  Credit risk: Alice/Bob bear bank risk pre-settlement; final leg
  settles in PUBLIC money (reserves) → singleness preserved.

(B) STABLECOIN
  - Alice's wallet -1 token; Bob's wallet +1 token (on-chain, instant)
  - NO central-bank leg. The "dollar" never touches public money.
  - Credit risk: Bob now holds a claim on the ISSUER's reserves.
    If reserves wobble, the token can trade below $1 (see USDC $0.88).

(C) RETAIL CBDC
  - Alice's CBDC balance -$1; Bob's +$1
  - This IS a central-bank liability moving directly. No intermediary
    credit risk at all; settlement asset and money are the same thing.
```

The punchline: in (A) the risky private claims are extinguished and replaced by public-money settlement; in (B) the private claim *is* the money and travels with its credit risk; in (C) there is no private credit risk to begin with. Same $1, three different places for the risk to sit.

REAL-WORLD EXAMPLE

In March 2023 roughly $3.3bn of Circle's USDC reserves were stuck at the failing Silicon Valley Bank. Holders could no longer be sure each USDC was redeemable for a full dollar *right now*, so the token fell to ~$0.88 before recovering once the US guaranteed SVB deposits. That episode is singleness of money breaking in real time: for a weekend, a USDC "dollar" and a bank-deposit "dollar" were not interchangeable at par.

BEST IN CLASS

The gold standard for digital money is to settle private-money transfers in public money wherever possible, exactly as the correspondent banking system does via central-bank reserves. Tokenized-deposit networks and wholesale CBDC (Chapters 05–06) are explicit attempts to carry this discipline onto DLT — keeping the final settlement leg in a central-bank liability so that singleness and par are preserved by design rather than by the issuer's good behavior.

04 · STABLECOINS

Mechanics, Reserves & Risk

A stablecoin is a privately issued token designed to hold a fixed value, almost always one US dollar. Behind that simple promise sit three very different engineering designs, a reserve portfolio that is really a money-market fund in disguise, an arbitrage mechanism that holds the peg, and a regulatory regime that arrived only after several violent failures. This chapter takes each apart with numbers.

The three designs

> Stablecoin — a token engineered to maintain a stable value (typically a 1:1 peg to a fiat currency) rather than to float freely like a cryptocurrency.

1. Fiat-reserve-backed — each token is backed by off-chain reserves (cash, Treasury bills, repo). Redeem a token, get a dollar. Tether (USDT), with a supply above roughly $120bn, and Circle (USDC), in the ~$30–60bn range, are the giants.
2. Crypto-over-collateralized — each token is backed by *more than* a dollar of volatile crypto collateral locked in a smart contract. DAI from MakerDAO is the archetype.
3. Algorithmic — no (or inadequate) hard reserves; the peg is maintained by an algorithm that expands and contracts a paired token. TerraUSD (UST) was the infamous example.

Reserve composition — why T-bills and repo dominate

A fiat-backed issuer must hold reserves that are safe, liquid, and yield-bearing. Short-dated US Treasury bills and overnight repurchase agreements (repo) fit all three: they carry negligible credit risk, can be sold or mature within days, and pay roughly the policy interest rate. Long bonds would add duration risk (price falls when rates rise); uninsured bank deposits add bank-failure risk — precisely the exposure that sank USDC's peg in 2023. The modern large stablecoin reserve therefore looks like a conservative government money-market fund.

> Repo (repurchase agreement) — a very short-term secured loan: the issuer lends cash and receives high-quality securities (usually Treasuries) as collateral, to be bought back the next day at a slightly higher price. It is effectively an overnight, collateralized deposit.

Primary mint/redeem versus the secondary market

> Primary mechanism — authorized parties mint new tokens by delivering $1 of cash to the issuer, and redeem tokens by returning them for $1. This happens at the issuer, always at par.

> Secondary market — everyone else trades the token on exchanges, where its price floats on supply and demand and can drift above or below $1.

The peg is held by arbitrage linking the two. If the secondary price falls below $1 while primary redemption is open, arbitrageurs buy cheap tokens and redeem them at the issuer for a full dollar, pocketing the gap and shrinking supply until the price climbs back. The mechanism only works while redemption is *open and credible* — the critical caveat.

WORKED EXAMPLE

The peg-arbitrage trade. Suppose USDC trades at $0.97 on an exchange while primary redemption at Circle is open at $1.00.

```
Buy 1,000,000 USDC on the exchange:  1,000,000 x $0.97 = $970,000
Redeem 1,000,000 USDC at Circle:     1,000,000 x $1.00 = $1,000,000
Gross profit:                        $1,000,000 - $970,000 = $30,000
Return on capital deployed:          $30,000 / $970,000 = 3.09%
```

A ~3.1% riskless-looking return, earned in the time it takes redemption to clear, is enormous — so arbitrageurs pile in, buying pressure lifts the price, and the discount closes. Why it broke in March 2023: with ~$3.3bn of reserves frozen at SVB, redemption was no longer *credibly* at $1.00 — nobody knew if the dollar would arrive or when — so the arbitrage stopped being riskless. The trade only pays if the final redemption leg is certain; remove that certainty and the peg falls to ~$0.88 until the US guaranteed SVB's deposits and certainty returned.

WORKED EXAMPLE

Issuer float-yield economics. An issuer holds $50bn of reserves in T-bills and repo yielding about 5% annually.

```
Annual gross reserve income = $50,000,000,000 x 0.05 = $2,500,000,000
```

That is ~$2.5bn per year, and the issuer typically pays token holders *nothing* on the balance — the holder gets a stable dollar, the issuer keeps the yield. This "float" is the entire business model: revenue scales linearly with supply and with the level of interest rates. At $120bn of supply (Tether's scale) and 5%, gross float income is $120bn x 0.05 = $6bn/yr. It also explains the industry's acute sensitivity to rate cuts: halve rates to 2.5% and the $50bn book's income halves to $1.25bn.

WORKED EXAMPLE

Over-collateralization and liquidation for a crypto-backed stablecoin (DAI-style). A borrower wants to mint $10,000 of stablecoin at a required 150% collateralization ratio, with a liquidation threshold also at 150%.

```
Required collateral = $10,000 x 1.50 = $15,000 of crypto (e.g. ETH)
Suppose ETH = $3,000, so deposit 5 ETH ($15,000).

Collateral ratio = (ETH price x units) / debt
At $3,000:  (3,000 x 5) / 10,000 = 150%  → just at threshold

ETH falls to $2,800:
   (2,800 x 5) / 10,000 = 140%  → BELOW 150%: liquidation triggers
The protocol sells enough ETH to repay the $10,000 debt + penalty,
protecting the stablecoin's backing. Price at which it triggers:
   P x 5 = 10,000 x 1.50  →  P = 15,000 / 5 = $3,000 per ETH
```

Because the system always holds more collateral than debt and force-sells before the buffer is exhausted, each DAI stays backed by >$1 of assets even as crypto prices swing — the opposite of the algorithmic design below.

Depeg episodes and why they happen

Depegs cluster around three causes: reserve doubt (USDC/SVB — the backing might not be there), redemption friction (you cannot get your dollar out fast enough, so the secondary price gaps down), and reflexive collapse (algorithmic designs). The UST catastrophe is the pure third case.

REAL-WORLD EXAMPLE

In May 2022 TerraUSD (UST) held its $1 peg by letting holders always swap 1 UST for $1 worth of its sister token LUNA, minted on demand. When confidence cracked and UST slipped below $1, arbitrageurs swapped UST for freshly minted LUNA, flooding the market with LUNA. LUNA's price collapsed, so ever more LUNA had to be minted to honor each $1 swap — a hyperinflationary "death spiral." Within days UST lost its peg entirely and roughly $40bn of combined value was wiped out. The lesson: a peg backed by a token you must print more of in a crisis is backed by nothing at the moment you need it.

Regulation — GENIUS Act and MiCA

Regulators responded to these failures by legislating what a payment stablecoin must be. The US GENIUS Act (2025) created a federal framework requiring payment stablecoins to be fully backed by high-quality liquid assets (cash and short-dated Treasuries), with clear redemption rights and reserve disclosure. The EU's MiCA regime imposes comparable rules: one-for-one reserves, strict safeguarding, and a legal right to redeem at par. Both regimes effectively outlaw the uncollateralized algorithmic model for anything marketed as a stable payment instrument.

> HQLA (high-quality liquid assets) — assets that can be sold quickly at little or no loss of value even in stress — chiefly cash and short-dated government securities. Reserve rules force stablecoin backing toward HQLA so redemption stays credible in a panic.

BEST IN CLASS

The best fiat-backed stablecoins converge on a simple formula: reserves held overwhelmingly in short-dated Treasuries and overnight repo (minimizing duration and bank-deposit risk), bankruptcy-remote custody so holders have a clean claim, independent attestations or audits of reserves, and a legally guaranteed par redemption under GENIUS/MiCA. That combination is precisely what USDC lacked on the SVB weekend (too much uninsured bank cash) and what UST never had at all (no hard reserves). Judge a stablecoin by how closely it matches this formula.

05 · CENTRAL BANK DIGITAL CURRENCIES

Wholesale & Retail CBDC

A central-bank digital currency is the simplest instrument in this book to define and the most consequential to design. It is public money in digital form — a direct claim on the central bank — but every design choice, from holding limits to privacy, reshapes the banking system around it. This chapter separates the retail and wholesale varieties and prices out their benefits.

What a CBDC is

> CBDC (central bank digital currency) — a digital form of a nation's sovereign money that is a *direct liability of the central bank*, exactly like physical cash but in electronic form. Unlike a bank deposit or a stablecoin, it carries no private-issuer credit risk.

Roughly 130 countries, representing the overwhelming majority of world GDP, are now exploring CBDCs according to central-bank trackers. The reasons vary — payment-system resilience, financial inclusion, countering private stablecoins, and modernizing cross-border settlement — but the common thread is a desire to keep public money relevant as cash use declines.

Retail versus wholesale

> Retail CBDC — central-bank digital money that ordinary households and businesses can hold and spend directly, a digital complement to cash. China's e-CNY pilot, with hundreds of millions of wallets, is the largest live example; the ECB's digital euro is in its preparation phase.

> Wholesale CBDC — central-bank digital money restricted to financial institutions for interbank and securities settlement — essentially a tokenized, programmable version of the reserve balances banks already hold.

Retail CBDC is politically charged (privacy, bank disintermediation) and technically ambitious. Wholesale CBDC is less controversial because it changes the *plumbing* among banks, not the public's relationship with money — which is why much of the near-term institutional action is wholesale.

The two-tier distribution model

> Two-tier model — the central bank issues the CBDC and operates the core ledger, but private intermediaries (banks, payment firms) distribute wallets, onboard customers, and provide services. The central bank stays out of retail customer relationships.

This mirrors how physical cash already works: the central bank prints it, commercial banks distribute it. Preserving the two-tier structure is a deliberate choice to avoid turning the central bank into a giant retail bank and to keep private-sector innovation in the loop.

Key design choices

- Account-based vs token-based — is access tied to a verified identity/account, or to control of a cryptographic token (more cash-like, more private)?
- Intermediated — distributed through banks (the two-tier model) rather than direct central-bank accounts for all.
- Offline capability — can it work without connectivity, like cash, for resilience and inclusion?
- Holding limits — a cap on how much any individual can hold, to limit bank disintermediation.
- Remuneration — does it pay interest? Usually proposed at zero to keep it a means of payment, not a savings vehicle.
- Programmability — can payments carry conditional logic (e.g. automatic tax, escrow)?
- Privacy — how much transaction data the state can see; the central tension of retail CBDC design.

Wholesale CBDC for interbank and cross-border settlement

The clearest institutional prize is cross-border payments, today routed through slow, liquidity-hungry correspondent banking.

> Correspondent banking (nostro/vostro) — to pay across borders, a bank holds pre-funded accounts at foreign banks ("nostro" = our account held there; "vostro" = your account held here). Payments hop through these accounts, each hop adding time, fees, and *trapped liquidity* — cash parked abroad doing nothing.

> Multi-CBDC bridge — a shared platform on which several central banks issue their wholesale CBDCs so banks can swap and settle directly in central-bank money across currencies, bypassing correspondent chains. BIS Project mBridge is the leading example.

WORKED EXAMPLE

Wholesale-CBDC cross-border payment versus correspondent banking. A bank must move the equivalent of $100m across borders.

```
CORRESPONDENT ROUTE
  Hops: originator → correspondent → beneficiary bank (2-3 intermediaries)
  Settlement time: ~1-3 business days (T+2 typical)
  Trapped liquidity: must PRE-FUND nostro accounts to transact.
    Say 10 corridors x $100m pre-funded = $1,000,000,000 sitting idle.
  Cost of that idle cash at 5% = $1,000,000,000 x 0.05 = $50,000,000/yr

MULTI-CBDC BRIDGE ROUTE (mBridge-style)
  Hops: direct atomic swap in central-bank money, 1 step
  Settlement time: near-instant (seconds), 24/7
  Trapped liquidity: little or none → free up ~$1,000,000,000.
    Annual carry saved at 5% = ~$50,000,000.
```

The headline saving is not the per-transaction fee; it is releasing roughly $1bn of pre-funded liquidity and ~$50m/yr of carrying cost, plus cutting settlement from days to seconds and removing counterparty risk during the wait.

WORKED EXAMPLE

Retail holding-limit and disintermediation. Suppose a digital euro carries a €3,000 holding cap and 10 million citizens each shift the full cap out of bank deposits into CBDC.

```
   Deposits migrating = 10,000,000 x EUR 3,000 = EUR 30,000,000,000
```

That is €30bn of funding leaving commercial banks for the central bank's balance sheet. Banks must replace it with pricier wholesale funding or shrink lending, so the cap is a direct lever on disintermediation: a €10,000 cap on the same 10 million people would move €100bn; a €500 cap just €5bn. This is exactly why holding limits exist — they bound how much deposit funding a retail CBDC can drain from the banking system in a stress event or bank run.

REAL-WORLD EXAMPLE

China's e-CNY is the largest retail CBDC pilot, with hundreds of millions of wallets, offline payment support, and a strictly two-tier model in which commercial banks distribute wallets. Even at that scale, balances held are deliberately small relative to total deposits — a real-world demonstration that design choices (low/zero remuneration, caps, positioning as a cash substitute) keep a retail CBDC from cannibalizing bank deposits.

BEST IN CLASS

The emerging consensus design is two-tier, intermediated, token-and-account hybrid, zero-remunerated, holding-capped retail CBDC for the public, paired with a programmable wholesale CBDC on multi-CBDC bridges (mBridge) for institutions. This keeps banks in the distribution loop, caps disintermediation risk, preserves a cash-like privacy option offline, and captures the large cross-border liquidity savings where they are easiest to realize — the wholesale layer — while moving cautiously on the politically sensitive retail layer.

06 · TOKENIZED DEPOSITS & FUNDS

Tokenized Deposits & Tokenized Money-Market Funds

The final foundational instruments bring tokenization inside the regulated banking and asset-management system. A tokenized deposit keeps your money a bank deposit — redeemable at par, inside deposit-insurance and supervision — while making it move at the speed and programmability of a token. Tokenized money-market funds go one step further, turning on-chain cash into a yield-bearing instrument you can subscribe to and redeem intraday.

What a tokenized deposit is

> Tokenized deposit (deposit token) — a commercial-bank deposit represented as a token on a ledger. It remains a legal bank deposit — a claim on the bank, redeemable 1:1 for ordinary deposits — and stays entirely within the regulated banking perimeter. The token is a new *form factor* for an existing deposit, not a new instrument.

This is the crucial distinction from a stablecoin. A stablecoin is a claim on a private issuer's reserve pool, sitting *outside* the banking system; a tokenized deposit is the same insured, supervised bank deposit you already understand, now able to settle on-chain 24/7 with programmable logic. Because it settles ultimately in central-bank reserves (Chapter 03), it preserves the singleness of money that stablecoins can jeopardize.

Feature	Tokenized deposit	Stablecoin
Legal nature	Bank deposit (claim on bank)	Claim on private issuer's reserves
Inside banking perimeter?	Yes	No
Deposit insurance	Yes (up to limit)	No
Final settlement asset	Central-bank reserves	Issuer reserves / itself
Redeemable at par 1:1?	Yes, by banking law	By issuer promise only
Who issues	Regulated commercial bank	Stablecoin company

24/7 programmable settlement

Today, wholesale bank payments run on business-hours rails with hard daily cut-off times; money sent after cut-off waits until the next business day. A deposit token settles continuously, instantly, and atomically, enabling delivery-versus-payment and intraday collateral movement that the legacy wire window cannot. The benefit is measured in *trapped overnight balances released*.

REAL-WORLD EXAMPLE

JPMorgan's Kinexys platform (formerly Onyx), which operates the JPM Coin deposit-token system, processes very large daily volumes of intraday programmable payments between the bank's institutional clients and moves tokenized collateral on-chain — for example, shifting money-market-fund shares as collateral in minutes rather than via end-of-day batch. It is the leading live demonstration that a deposit token can carry real institutional cash-management traffic inside a regulated bank.

WORKED EXAMPLE

Trapped overnight balance saved by 24/7 atomic payment. A corporate treasury must fund a $200m obligation that becomes due at 7:00 pm, two hours after the 5:00 pm wire cut-off.

```
CUT-OFF-BOUND WIRE
  Miss the 5pm window → cannot pay until next business day.
  Treasury must PRE-FUND today to be safe: park $200,000,000 overnight.
  Overnight carry forgone at ~5% annual (1 night = 1/360):
     $200,000,000 x 0.05 x (1/360) = $27,777.78 for ONE night
  Over 250 business nights/yr of such buffers:
     $27,777.78 x 250 = $6,944,444 per year

24/7 ATOMIC DEPOSIT-TOKEN PAYMENT
  Pay at 7pm exactly when due; nothing pre-funded overnight.
  Trapped balance released: $200,000,000
  Annual carry recaptured: ~$6.9m on this single recurring flow.
```

The value of continuous atomic settlement is the elimination of defensive pre-funding: the treasury pays at the moment of need instead of parking idle cash to beat an arbitrary cut-off.

Tokenized money-market funds and on-chain cash management

> Tokenized money-market fund (tokenized MMF) — shares of a regulated money-market fund — a fund that invests in short-dated government and cash-like instruments — issued as tokens on a ledger. Each token is a fund share that accrues the fund's yield and can be subscribed to or redeemed on-chain, often around the clock.

This gives institutions yield-bearing on-chain cash: unlike a stablecoin (which typically pays the holder nothing), a tokenized MMF passes the underlying T-bill yield through to the holder while remaining highly liquid. It is the natural on-chain home for idle treasury cash.

REAL-WORLD EXAMPLE

BlackRock's BUIDL is an on-chain money fund holding cash and short-term US Treasuries, with shares issued as tokens that accrue daily yield — rapidly becoming a benchmark for institutional on-chain cash. Franklin Templeton's tokenized money fund was an early mover, recording fund-share ownership directly on a public blockchain. Both show large, regulated asset managers using tokenization to deliver T-bill yield in a 24/7, transferable form.

WORKED EXAMPLE

Tokenized-MMF subscription, intraday redemption, and accrued yield. An institution subscribes $10,000,000 into a tokenized MMF yielding 5.00% annualized, then redeems after 30 days.

```
Subscribe: $10,000,000 → receives 10,000,000 tokens at $1.00 NAV
Daily yield accrual (actual/360 convention):
   daily rate = 0.05 / 360 = 0.0001388...
   per day on $10m = $10,000,000 x 0.0001388... = $1,388.89/day
Over 30 days:
   $1,388.89 x 30 = $41,666.67 accrued yield
Redeem on day 30 (intraday, on-chain):
   principal + yield = $10,000,000 + $41,666.67 = $10,041,666.67
```

Two things a stablecoin cannot match: the holder *earned* ~$41.7k of yield over the month, and the redemption settled intraday on-chain rather than waiting for a next-day fund-administrator batch. Idle cash that would have sat in a non-yielding token instead compounded at the T-bill rate while staying liquid.

Yield-bearing on-chain cash — the convergence

Put the pieces together and a new cash stack emerges: tokenized deposits for par-stable, insured *payment* balances inside the bank; tokenized MMFs for yield-bearing *investment* cash; and (prospectively) wholesale CBDC as the ultimate public-money settlement layer beneath both. Each solves the problem stablecoins only approximate — par settlement, regulatory protection, and passed-through yield — while keeping the 24/7 programmability that made tokenization attractive in the first place.

BEST IN CLASS

The institutional frontier is a bank-issued deposit token for atomic payments and intraday collateral (the Kinexys/JPM Coin model) wired to a tokenized government money-market fund for yield (the BUIDL / Franklin Templeton model), so a treasurer can hold yield-bearing tokenized MMF shares, redeem them on-chain in seconds into par-stable deposit tokens exactly when a payment is due, and settle atomically — all inside the regulated perimeter, with no cut-off windows and no idle, non-yielding cash. That is tokenized cash management done right: par where you need stability, yield where you need return, and 24/7 programmability throughout.

07 · REAL-WORLD ASSETS

Tokenizing Real-World Assets — Bonds, Funds, Private Credit & Treasuries

Tokenization reaches institutional scale not through speculative coins but through ordinary financial assets — bonds, fund units, loans, real estate — represented as programmable tokens on a shared ledger. This chapter follows such a security from origination to redemption, surveys the asset classes already live, and separates the genuine benefits from the honest limits that every desk must price in.

7.1 What RWA tokenization actually means

A real-world asset (RWA) is any asset whose economic value and legal existence sit off-chain — a government bond, a share in a money-market fund, a mortgage, a warehouse of copper. Tokenizing it means issuing a digital token on a ledger that represents a legal claim on that asset, so that holding, transferring, and servicing the claim happen on-chain while the asset itself (and its law) remains anchored in the real world.

> Tokenization — the process of issuing a transferable digital token on a distributed ledger that represents ownership of, or a claim on, an underlying asset, such that the token becomes the authoritative record of that claim.

> Real-world asset (RWA) — a traditional, off-chain financial or physical asset (bond, loan, fund unit, property, commodity) whose on-chain token form is governed by the same contracts and law as the underlying.

> Legal wrapper — the contractual and corporate structure (a trust, an SPV, a note deed, a fund prospectus) that legally binds the on-chain token to the off-chain asset so a token holder has an enforceable claim.

The crucial distinction from the first crypto wave is that an RWA token is not the asset; it is a representation whose authority depends entirely on the legal wrapper. A tokenized bond is enforceable only because a note deed says the registered token holder is the creditor. Strip the wrapper away and the token is a number. This is why the lawyers, not the coders, decide whether a tokenization project is real.

7.2 The lifecycle of a tokenized security

Every tokenized security moves through the same six stages. Understanding them is the core competence of this subject.

1. Origination — the issuer and arranger structure the instrument: tenor, coupon, covenants, and the legal wrapper that makes the token a claim.
2. Issuance / mint — tokens are created ("minted") on the ledger against the deed; the primary allocation is recorded natively on-chain rather than in a separate registrar's book.
3. Distribution — tokens are delivered to primary investors, ideally on a delivery-versus-payment basis so cash and token swap atomically (Chapter 08).
4. Servicing — coupons, dividends, and corporate actions are computed and paid by smart contract pro-rata to the holders recorded at a snapshot.
5. Secondary trading — holders transfer tokens peer-to-peer or on a regulated venue, with the ledger updating ownership in near real time.
6. Redemption — at maturity the principal is paid and the tokens are burned (destroyed), closing the register.

> Mint / burn — to *mint* is to create new tokens on the ledger (at issuance); to *burn* is to permanently destroy them (at redemption or when units are retired), so the token supply always equals the outstanding claim.

> Paying agent — the institution that, in a conventional bond, receives cash from the issuer and distributes coupons and principal to holders; in a tokenized bond much of this role is automated by the servicing smart contract.

7.3 Asset classes being tokenized

Asset class	What is tokenized	Why it fits	Live examples
Government bonds	Sovereign & supranational debt	Standardized, high-grade, clear law	World Bank bond-i; EIB; Hong Kong green bonds
Corporate bonds	Senior notes, commercial paper	Automated coupons, broader distribution	Siemens ~€60m (2023)
Money-market funds	Fund units / shares	24/7 subscription, use as collateral	BlackRock BUIDL; Franklin Templeton
Private credit	Direct loans, loan pools	Fractionalization, servicing transparency	Various private-credit platforms
Private equity / funds	LP interests	Smaller tickets, secondary liquidity	Feeder-fund pilots
Real estate	SPV equity in a property	Fractional ownership, lower minimums	Commercial-property pilots
Commodities	Allocated gold, metals	Provable backing, divisibility	Tokenized gold programs

7.4 Named examples

REAL-WORLD EXAMPLE

In August 2018 the World Bank issued bond-i ("blockchain operated new debt instrument"), widely regarded as the first bond created, allocated, and managed end-to-end on a blockchain, raising approximately A$110m over a two-year tenor on a private Ethereum-based ledger arranged by CBA. The European Investment Bank (EIB) followed with a series of digital bonds from 2021, including a ~€100m issuance on the public Ethereum chain and later deals on private and central-bank-money rails. Siemens issued an approximately €60m one-year digital bond in 2023 under Germany's electronic-securities law (eWpG), selling directly to investors without the usual central paper certificate or clearing intermediary. Hong Kong's government sold an approximately HK$800m tokenized green bond in 2023 and returned with a multi-currency tokenized issue of roughly HK$6bn, demonstrating sovereign-scale, multi-currency tokenized debt.

REAL-WORLD EXAMPLE

BlackRock BUIDL (the USD Institutional Digital Liquidity Fund), launched March 2024 on Ethereum with Securitize as transfer agent, tokenizes a short-dated US-Treasury and repo money fund. It grew past roughly $500m within months and became a reference point for using a tokenized money-market fund as on-chain, yield-bearing collateral. More broadly, the tokenized-US-Treasury market expanded from around $100m to several billion dollars across 2023–2024.

7.5 Issuance economics — tokenized vs conventional

WORKED EXAMPLE

Consider a $200m three-year senior bond.

```
Conventional issuance (indicative, per deal)
  Paying-agent & registrar fees .......  $120,000
  Clearing / settlement fees ..........   $90,000
  Issuer admin & reconciliation .......   $60,000
  Settlement lag (T+2) funding cost* ..   $44,000
  TOTAL (illustrative) ................  $314,000

  *Principal tied up 2 extra days:
   $200m x 4.0% x (2/360) = $44,444

Tokenized issuance (indicative)
  Platform / registrar-on-ledger ......   $70,000
  Settlement (atomic, same day) .......   $20,000
  Issuer admin (automated servicing) ..   $15,000
  Settlement lag funding cost .........        $0
  TOTAL (illustrative) ................  $105,000

  Indicative saving: $314,000 - $105,000 = $209,000 (~66%)
```

The headline cash saving is modest against a $200m principal, but the structural wins — atomic settlement, a single golden register, and automated coupons — compound across a programme of many deals and remove reconciliation headcount.

7.6 Smart-contract coupon flow

WORKED EXAMPLE

A tokenized bond pays a 5.0% annual coupon, semi-annual, on $200m face, split into 200,000 tokens of $1,000 face each.

```
Semi-annual coupon pool = $200m x 5.0% x (1/2) = $5,000,000
Per-token coupon        = $5,000,000 / 200,000  = $25.00

At the record-date snapshot the contract reads the register:
  Holder A: 40,000 tokens → 40,000 x $25 = $1,000,000
  Holder B: 12,500 tokens → 12,500 x $25 =   $312,500
  Holder C:    500 tokens →    500 x $25 =    $12,500
Issuer funds $5,000,000 in the settlement asset; the
contract disburses pro-rata in one atomic batch. No
paying agent, no manual pro-ration, no failed payments.
```

7.7 Fractionalization reach

WORKED EXAMPLE

A property SPV worth $50m conventionally sold in $5m tickets reaches at most 10 investors. Tokenized into units of $1,000: $50m / $1,000 = 50,000 units, addressable by tens of thousands of investors. Lowering the minimum from $5m to $1,000 widens the eligible-investor universe by a factor of 5,000 — the single clearest commercial argument for tokenizing illiquid assets.

7.8 Benefits and honest limits

Benefits: fractionalization (above), faster and atomic settlement, automated servicing, a single transparent golden source of holdings, and new distribution channels that reach investors a syndicate desk never could.

Honest limits: the token is only as strong as its legal wrapper; secondary liquidity is often thin because a fractional pool of buyers must actually exist; and off-chain enforceability means that if a borrower defaults, you still go to a court and a receiver, not to a smart contract. Tokenization improves plumbing; it does not repeal credit risk or property law.

BEST IN CLASS

A best-in-class RWA programme treats the token as the authoritative register (not a shadow of a paper book), settles primary allocation on a true DvP basis, automates 100% of coupon and corporate-action servicing by contract, and — critically — publishes a legal wrapper whose governing law makes the registered token holder the enforceable creditor. Siemens' eWpG issuance and the Hong Kong multi-currency programme are the reference points: legally native digital securities, not paper bonds with a blockchain bolted on.

08 · MARKET INFRASTRUCTURE

Market Infrastructure — Settlement, DvP, Atomic Settlement & Smart Contracts

A tokenized security is only as good as the plumbing that moves it. This chapter explains how tokenized markets clear and settle — delivery-versus-payment, atomic settlement, the role of smart contracts and oracles, custody and key management, interoperability, and the single most consequential design choice of all: what the cash leg actually settles in.

8.1 Clearing and settlement, restated

Clearing is everything between trade and settlement — confirmation, netting, margining. Settlement is the final exchange of asset for cash. In conventional markets these are separate, sequential, and slow (T+1 or T+2). On a shared ledger they can collapse into a single event.

> Settlement finality — the moment a transfer becomes irrevocable and unconditional in law, so it cannot be unwound even if a party later fails.

> Settlement cycle (T+n) — the number of business days after trade date (T) on which settlement occurs; US equities moved to T+1 in 2024, but much of fixed income and FX still runs T+2.

8.2 DvP, PvP, and atomic settlement

> Delivery-versus-payment (DvP) — a settlement mechanism in which the transfer of the security occurs if, and only if, the corresponding payment occurs — eliminating the risk that you deliver the asset and never get paid.

> Payment-versus-payment (PvP) — the same principle for two cash legs, as in FX: one currency is delivered if and only if the other is, eliminating settlement risk between the two.

> Atomic settlement — both legs of a transaction settle in a single indivisible ledger operation — either everything completes together or nothing does — making DvP/PvP mechanically guaranteed rather than operationally promised.

Atomic settlement is the headline capability of tokenized markets. In a traditional DvP chain, the "if and only if" is enforced by a sequence of messages between custodians and a CSD, with timing gaps in which one leg

has moved and the other has not. On a unified ledger the swap is one state transition, so T+0 — same-moment settlement — becomes the default, not an aspiration.

> Herstatt / principal risk — the risk, named after Bankhaus Herstatt (failed 1974), that you irrevocably pay or deliver one leg and your counterparty fails before delivering the other, so you lose the full principal, not just a mark-to-market move.

8.3 Worked example — the capital and risk saving of atomic DvP

WORKED EXAMPLE

A dealer buys a $500m bond block, conventionally settling T+2.

```
Principal (Herstatt) exposure during settlement window
  T+2 process: full $500m at risk for up to 2 days.
  Expected principal loss (illustrative):
    $500m x 0.10% counterparty default prob
          x 100% loss-given-default (principal)
    = $500,000 expected exposure removed by atomic DvP.

Funding of settlement margin
  To cover that open exposure a CCP/counterparty posts
  initial margin, say 2% of notional:
    $500m x 2% = $10,000,000 of margin tied up for 2 days.
  Funding cost at 4.5% for 2/360 days:
    $10,000,000 x 4.5% x (2/360) = $2,500.
  Over ~250 settlement days/yr at similar scale this
  recurring margin funding runs into millions.

Atomic DvP (T+0): principal exposure ~ $0 (legs are
indivisible), and the 2-day settlement margin is freed.
```

WORKED EXAMPLE

Intraday timeline comparison for the same trade:

```
Conventional (T+2)
  Day 0  09:30  Trade agreed
  Day 0  17:00  Confirmation/affirmation
  Day 1         Netting, margin posted, instructions
  Day 2  ~15:00 DvP at CSD → finality
  ⇒ ~53 hours of open principal/counterparty exposure

Tokenized (atomic, T+0)
  09:30:00.0  Trade agreed
  09:30:02.5  Single ledger op: token + cash swap
  09:30:02.5  Finality
  ⇒ ~2.5 seconds of exposure
```

8.4 Smart contracts, oracles, and oracle risk

> Smart contract — code deployed on a ledger that automatically executes agreed logic (transfers, coupon payments, collateral top-ups) when defined conditions are met, without a manual intermediary.

> Oracle — a service that brings trusted off-chain data (a price, an interest rate, a credit event, a KYC status) onto the ledger so smart contracts can act on real-world facts.

> Oracle risk — the danger that the off-chain data feeding a contract is wrong, stale, or manipulated, causing the contract to execute correctly on bad inputs (e.g. liquidating collateral on a spoofed price).

Because a smart contract executes deterministically, an oracle is both indispensable and the system's softest attack surface. Best practice uses multiple independent oracles with medianized prices and sanity bounds, and circuit-breakers that halt execution when a feed deviates implausibly.

8.5 Custody, wallets, keys, and MPC

> Private key — the secret cryptographic value that authorizes transfers of a token; whoever controls the key controls the asset — "not your keys, not your coins."

> Multi-party computation (MPC) — a technique that splits a private key into shares held by several parties so that signing requires a quorum and no single party ever holds the whole key, removing the single point of failure.

Institutional custody rarely uses a single key. It uses MPC or multisignature schemes so that a transfer needs, say, three of five authorized shares, combined with hardware security modules, withdrawal allow-lists, and policy engines. Key management — not consensus algorithms — is where real-world losses happen.

8.6 Interoperability and bridges

> Cross-chain bridge — infrastructure that moves or represents a token's value from one ledger to another; historically a major source of exploits because the locked assets form a concentrated honeypot.

No single chain will host all assets, so interoperability matters. Institutional designs favour a small number of permissioned ledgers linked by controlled, auditable connections or shared settlement layers, rather than public bridges with weak governance.

8.7 The settlement-asset choice

When the token leg settles, what is the cash leg? The answer determines the credit risk embedded in every trade.

Settlement asset	What it is	Credit risk	Example rail
Stablecoin	Token backed by issuer reserves	Issuer & reserve-quality risk	Regulated USD stablecoins
Tokenized deposit	Commercial-bank money on-chain	Bank credit risk	JPM Kinexys deposit token
Wholesale CBDC	Central-bank money on-chain	Effectively risk-free	Fnality; BIS Agorá

> Wholesale CBDC — a tokenized form of central-bank reserves used by financial institutions to settle large-value transactions, carrying the credit quality of the central bank itself.

Settling in central-bank money matters because it removes settlement-asset credit risk entirely: the cash leg cannot default. This is why Fnality (wholesale settlement in central-bank money) and the BIS Project Agorá vision are treated as the gold standard, with tokenized deposits a strong commercial-bank alternative and stablecoins acceptable only where reserve quality and redemption are tightly regulated.

REAL-WORLD EXAMPLE

SIX SDX in Switzerland operates a fully regulated digital exchange and CSD where tokenized securities settle DvP against tokenized Swiss-franc cash, and DTCC Project Ion demonstrated a resilient T+0/ atomic settlement platform running in parallel with the traditional US settlement system — showing that atomic DvP is a production capability, not a whiteboard idea.

BEST IN CLASS

The best-in-class market settles atomic DvP in central-bank money (or a fully-reserved regulated equivalent) on a permissioned ledger, uses MPC custody with policy engines, feeds contracts from medianized multi-oracle price sources with circuit-breakers, and avoids public cross-chain bridges in favour of a shared or interoperable regulated settlement layer. Fnality-style CeBM settlement combined with SDX-style regulated DvP is the template every wholesale project is converging on.

09 · TREASURY & FUNDING ON-CHAIN

Treasury & Funding On-Chain — Repo, Collateral Mobility & Intraday Liquidity

Tokenization's biggest near-term prize is not retail investing but the wholesale treasury and funding function — repo, collateral, and liquidity management. When collateral moves in minutes and repo can be borrowed by the hour, the economics of running a bank or dealer balance sheet change materially. This chapter quantifies that change.

9.1 Why treasury is the first battleground

The treasury desk's job is to fund the balance sheet at least cost while keeping enough high-quality liquid assets to survive stress. Its enemies are settlement lag and collateral fragmentation — assets stuck in the wrong account, in the wrong time zone, overnight. Tokenization attacks both by making settlement instant and continuous and by letting a single pool of collateral move freely.

> Repo (repurchase agreement) — a secured loan: a borrower sells a security for cash and agrees to buy it back later at a slightly higher price, the difference being the interest (the repo rate).

> Collateral mobility — the speed and ease with which a security can be moved from where it sits to where it is needed as collateral; traditionally constrained by custodian cut-offs and T+1 settlement.

> Triparty — an arrangement where a neutral agent holds and manages collateral between a borrower and lender, handling selection, valuation, and substitution.

9.2 24/7 instant settlement and intraday repo

On a shared ledger, settlement is not bound to a clearing-house cut-off. Collateral and cash move whenever needed, including overnight and on weekends, and repo can be struck for a few hours instead of overnight — intraday repo. Combined with automated margin (a contract that marks exposure and calls or returns collateral continuously), the treasury desk manages a single programmable pool rather than a patchwork of accounts.

REAL-WORLD EXAMPLE

Broadridge Distributed Ledger Repo (DLR) settles repo on a private ledger with atomic DvP and has reached well over $1tn of repo volume per month, enabling intraday and even hour-by-hour repo that conventional plumbing cannot support. JPMorgan's Tokenized Collateral Network (part of the Kinexys/Onyx franchise) let a client pledge BlackRock money-market-fund shares as tokenized collateral, moving them intraday to a counterparty (Barclays) in a landmark 2023 demonstration. HQLAx provides a collateral-mobility layer that lets institutions transfer ownership of baskets of securities across custodians precisely and instantly, and Fnality provides the central-bank-money cash leg for such transfers.

9.3 Worked example — the funding saving from intraday repo

WORKED EXAMPLE

A dealer needs $1bn of cash to cover a 4-hour intraday funding gap; the repo rate is 5.0%.

```
Conventional: must borrow overnight (no shorter tenor)
  Interest = $1bn x 5.0% x (1/360) = $138,889 for 1 day.

Intraday repo: borrow for exactly 4 hours (4/24 of a day)
  Interest = $138,889 x (4/24) = $23,148.

  Saving per event = $138,889 - $23,148 = $115,741.

If this gap recurs on ~200 business days/yr:
  Annual saving ~ $115,741 x 200 = ~$23.1m.
```

Paying for only the hours you actually borrow is the single clearest economic case for on-chain funding.

9.4 Worked example — liquidity-buffer saving from faster collateral mobility

WORKED EXAMPLE

A desk routinely sends out $2bn of collateral and, because it returns T+1, must hold a duplicate intraday buffer to cover the day the collateral is "in flight."

```
Conventional: collateral out and unavailable ~1 day.
  Required duplicate HQLA buffer = $2bn.
  Carrying cost of that buffer (opportunity cost of
  holding low-yield HQLA vs funding), say 1.5% spread:
    $2bn x 1.5% = $30,000,000 per year.

Tokenized: collateral returns in ~minutes, so the
in-flight window is near zero.
  Needed buffer to cover in-flight collateral ~ $0.
  Buffer released ~ $2bn.
  Annual carry saved ~ $30,000,000.
```

Faster mobility does not just speed operations; it shrinks the balance sheet a bank must carry to be safe.

9.5 Collateral optimization across a unified pool

> Collateral optimization — allocating the cheapest-to-deliver eligible collateral against each obligation while keeping the most valuable assets free, maximized when all collateral sits in one mobile pool rather than siloed accounts.

When every eligible asset lives in one tokenized pool with instant mobility, an optimization engine can continuously re-allocate: pledge the lowest-quality eligible bond here, free the on-the-run Treasury to be lent out there, and substitute in seconds when a margin call lands. The gain is both a lower funding bill and a smaller required inventory of premium collateral.

Treasury function	Conventional	Tokenized / on-chain
Repo tenor granularity	Overnight minimum	Hourly / intraday
Collateral move time	Up to T+1	Minutes
Operating hours	Clearing-house hours	24/7
Margin process	Periodic, manual	Continuous, automated by contract
Collateral view	Fragmented accounts	Single unified pool

BEST IN CLASS

The best-in-class treasury runs a single programmable collateral pool that mobilizes in minutes, funds intraday with hour-granular repo on a Broadridge-DLR-style ledger, settles the cash leg in central-bank money (Fnality) to remove settlement-asset credit risk, and lets automated triparty margin continuously optimize cheapest-to-deliver collateral across the whole book. JPMorgan's intraday pledge of tokenized MMF shares and HQLAx's instant ownership transfers are the live proof points that this operating model already works at scale.

10 · RISKS, CONTROLS & REGULATION

Risks, Controls & Regulation

Tokenization re-plumbs finance; it does not abolish risk, and it adds new failure modes of its own. An expert can name every risk, say exactly how it bites, and pair it with a concrete control — and can compute the capital a regulator will make a bank hold against these assets. This chapter builds that risk map and does the prudential arithmetic.

10.1 The real risk map

The dangers fall into technical, custody, data, settlement, counterparty, and governance buckets. The novice sees "crypto risk"; the expert sees a specific, controllable list.

> Smart-contract risk — the risk that a bug or logic flaw in the contract code is exploited, causing loss or incorrect execution; mitigated by audits, formal verification, and value caps.

> Stablecoin run risk — the risk that holders doubt a stablecoin's reserves and redeem en masse, forcing fire-sales that break the peg (as with the UST collapse in 2022).

> Governance / upgradeability risk — the risk that an administrative key or governance process can change, pause, or upgrade a contract — useful for fixes, dangerous if that control is captured or misused.

> FATF travel rule — the Financial Action Task Force requirement that transfers of value carry originator and beneficiary identity information, applied to crypto-asset transfers between regulated entities for AML purposes.

Risk	How it bites	Control
Smart-contract bug	Exploit drains or misroutes value	Independent audits, formal verification, value caps, bug bounties
Private-key loss / theft	Irreversible loss of the asset	MPC/multisig, HSMs, allow-lists, segregated keys
Oracle manipulation	Contract acts on spoofed price	Multiple medianized oracles, deviation circuit-breakers
Stablecoin run / reserve quality	Settlement asset breaks peg	CeBM or fully-reserved regulated cash; redemption rights
Settlement-asset credit risk	Cash leg issuer fails	Settle in central-bank money
Chain / provider concentration	One outage halts the market	Multi-chain resilience, fallback procedures
Governance key capture	Malicious upgrade or freeze	Timelocks, multi-party governance, transparency
Legal finality gap	On-chain transfer not final in law	Legal wrapper + statute (e.g. eWpG, DLT pilot regimes)
AML / sanctions	Illicit flows, enforcement risk	KYC at onboarding, travel-rule messaging, screening

10.2 Legal finality — is an on-chain transfer actually final?

A ledger entry is technically irreversible, but legal finality is a separate question answered by law, not code. Jurisdictions have responded with purpose-built regimes — Germany's eWpG electronic-securities law, Luxembourg's blockchain laws, and the EU's DLT Pilot Regime — that make the ledger entry the legally authoritative record. Without such a statute, a token transfer may be only a pointer to an off-chain registrar's book, and finality is uncertain.

10.3 The prudential treatment — Basel crypto-asset standard

> Basel crypto-asset standard — the Basel Committee's framework for how banks must hold capital against crypto-assets, splitting them into Group 1 (tokenized traditional assets and compliant stablecoins meeting strict conditions, treated broadly like the underlying) and Group 2 (everything else — unbacked cryptoassets and those failing the conditions), which can attract a punitive 1250% risk weight.

> Risk weight — the percentage of an exposure that counts as risk-weighted assets (RWA) for capital purposes; capital required = RWA × the bank's capital ratio.

WORKED EXAMPLE

A bank holds $100m of an asset. Assume an 8% minimum capital ratio.

```
Group 2 (1250% risk weight)
  RWA = $100m x 1250% = $1,250,000,000
  Capital required = $1.25bn x 8% = $100,000,000
  ⇒ ~$1 of capital for every $1 held (a dollar-for-
     dollar deduction in effect). Holding it is
     capital-prohibitive for a bank.

Group 1 tokenized government bond (treated like the
underlying sovereign; say 0% risk weight for a
qualifying high-grade sovereign)
  RWA = $100m x 0% = $0
  Capital required = $0.
  ⇒ The tokenization itself adds no capital charge;
     the asset is treated by its economic substance.
```

This is the single most important regulatory fact in the field: a compliant tokenized government bond is Group 1 and essentially free of extra capital cost, while a non-compliant crypto-asset is Group 2 and effectively un-investable for a regulated bank. Classification, not technology, decides the economics.

10.4 The regulatory regimes to know

> MiCA (Markets in Crypto-Assets) — the EU regulation (phasing in from 2024) that licenses crypto-asset issuers and service providers and imposes reserve, disclosure, and redemption rules on stablecoins (asset-referenced and e-money tokens).

> GENIUS Act — US federal legislation (2025) establishing a regulatory framework for payment stablecoins — reserve backing, redemption, and issuer licensing — bringing US dollar stablecoins into supervised scope.

Together, MiCA in Europe and the GENIUS Act in the US move stablecoins from the shadows into regulated, fully-reserved instruments — which is precisely what makes them acceptable settlement assets for institutions and what pushes a stablecoin toward Basel Group 1 eligibility.

> Banks' crypto-asset exposures must be subject to a conservative prudential treatment, with Group 2 exposures capped and risk-weighted at 1250%, reflecting the higher risks involved.
>
> — Basel Committee on Banking Supervision, crypto-asset standard

BEST IN CLASS

A best-in-class programme engineers every instrument to qualify as Basel Group 1, issues under a statute that confers legal finality (eWpG, the EU DLT Pilot Regime), uses only MiCA/GENIUS-compliant fully-reserved settlement assets, runs MPC custody with timelocked multi-party governance, and bakes travel-rule and sanctions screening into the token's transfer logic. The test is simple: could a conservative bank supervisor look at the structure and treat it like the underlying traditional asset? If yes, you have built it right.

11 · CAPSTONE

What "Best in Class" Looks Like — The Institutional Operating Model & the Road Ahead

This capstone assembles everything into the operating model a world-class institution actually runs, sets out the BIS "unified ledger" vision, gives a realistic adoption roadmap and the KPIs that measure progress, and ends with how an individual turns this knowledge into recognized expertise. It is the blueprint for going from laggard to leader.

11.1 The institutional tokenization operating model

Strip away the hype and a best-in-class model has seven pillars, each established in the preceding chapters:

1. Settlement in tokenized central-bank or commercial-bank money — the cash leg carries no (or minimal) credit risk.
2. Atomic DvP/PvP — both legs settle together or not at all; T+0 is the default.
3. A single programmable collateral pool — one mobile inventory, continuously optimized.
4. Interoperable permissioned ledgers — a small set of regulated ledgers linked by shared settlement, not public bridges.
5. Robust custody and key management — MPC/multisig, HSMs, policy engines.
6. Clear legal wrappers — statute-backed finality so the token is the enforceable claim.
7. Compliance by design — Basel Group 1 structuring, travel-rule and sanctions logic embedded in transfers.

11.2 The BIS "unified ledger" vision

> Unified ledger — the BIS concept of a common programmable platform on which tokenized central-bank money, commercial-bank money, and assets coexist, so that complex multi-step transactions settle atomically in one place rather than across fragmented systems.

REAL-WORLD EXAMPLE

Project Agorá is a BIS-led initiative with seven central banks and a large group of private banks exploring how tokenized commercial-bank deposits and wholesale central-bank money can settle together on a unified platform — the most ambitious expression of the unified-ledger vision. Project Pine (BIS Innovation Hub) explored how a central bank could implement monetary-policy tools (such as interest on reserves and asset facilities) using tokenized instruments and smart contracts. Alongside them, Partior (a bank-owned cross-border settlement network) and Fnality provide the live commercial rails that make the vision concrete.

11.3 A realistic adoption roadmap

Adoption will not be a big bang. The credible sequence, already visible in the market, is:

1. Wholesale first — repo, FX, and interbank settlement, where counterparties are sophisticated and the plumbing savings are largest.
2. Collateral and funds — tokenized money-market funds and collateral mobility (BUIDL, JPMorgan TCN, HQLAx).
3. Broad RWA — corporate and sovereign bonds at scale, then private credit, private funds, and real estate reaching broader investors.

Retail-facing, fully public RWA markets are the last stage, not the first, because they depend on all the wholesale plumbing and legal finality being solved beneath them.

11.4 The KPIs an institution tracks

KPI	What it measures	Leader target (illustrative)
Atomic settlement share	% of settlement value settling atomically	> 80%
Collateral mobility time	Median time to move collateral	< 5 minutes
On-chain AUM	Assets serviced natively on-ledger	Growing double-digit % of book
Cost per trade	Fully-loaded settlement + servicing cost	Falling year on year
Failed-trade rate	% of trades failing to settle on time	Near 0% (atomic removes fails)
Settlement-asset quality	% settled in CeBM / fully-reserved cash	> 90%

WORKED EXAMPLE

A dealer settling $500bn/yr cuts its failed-trade rate from 2% to near 0% via atomic DvP. If each fail costs ~5bp of the trade in funding penalties and buy-in risk:

```
Failed notional before = $500bn x 2% = $10bn
Cost of fails          = $10bn x 0.05% = $5,000,000/yr
Atomic DvP failed notional ~ $0  →  ~$5m/yr saved,
plus the capital freed from fail-related buffers.
```

The KPIs are not vanity metrics; each maps directly to money.

11.5 A laggard-to-leader maturity model

Stage	Settlement	Collateral	Custody / legal
1 — Laggard	T+2, manual DvP	Fragmented, T+1 moves	Paper registrar, no DLT law
2 — Pilot	Single tokenized bond, off-chain cash	One pledge demo	Custodial single-key, legal pilot
3 — Scaling	Atomic DvP, tokenized-deposit cash	Unified pool, intraday repo	MPC custody, statute-backed finality
4 — Leader	Atomic DvP in CeBM, T+0 default	Optimized pool, minutes mobility	MPC + governance timelocks, Group 1 everywhere

11.6 How an individual becomes a recognized expert

Expertise in this field is demonstrated by three things: fluent vocabulary, command of the canon, and the ability to run the numbers on any proposal (as this guide's worked examples train you to do).

- The vocabulary — DvP/PvP, atomic settlement, settlement finality, collateral mobility, CeBM vs tokenized deposit vs stablecoin, Basel Group 1/2, MPC, oracle risk, legal wrapper, unified ledger. Use each precisely.
- The canon — BIS Annual Economic Report chapters on the unified ledger and tokenization, and BIS Innovation Hub project reports (Agorá, Pine, mBridge, Helvetia); the Basel crypto-asset standard; MiCA and the GENIUS Act texts; and the primary platform documentation (DTCC, Broadridge DLR, JPMorgan Kinexys, Fnality, SIX SDX, HQLAx, BlackRock BUIDL/Securitize).
- The money-view thinkers — the "money view" school (Perry Mehrling and colleagues) for understanding settlement, liquidity, and the hierarchy of money, which is the intellectual bedrock beneath every settlement-asset debate.
- The habit — for any tokenization pitch, immediately ask: what settles the cash leg, is it atomic, is it Basel Group 1, who holds the keys, and what statute gives it finality? If you can answer those five, you are operating at expert level.

REAL-WORLD EXAMPLE

The clearest sign the model is real: by 2024 Broadridge DLR was settling more than $1tn/month of repo, BlackRock BUIDL had passed ~$500m of tokenized Treasuries, JPMorgan had pledged tokenized MMF shares as intraday collateral, Germany had live eWpG digital bonds, Hong Kong had sold multi-currency tokenized bonds near HK$6bn, and the BIS had convened Project Agorá. The leader's playbook is no longer theoretical — it is being assembled, deal by deal, in production.

BEST IN CLASS

The best-in-class institution settles atomic DvP in central-bank money, runs one optimized programmable collateral pool that mobilizes in minutes, issues every instrument as Basel Group 1 under a finality-conferring statute, secures assets with MPC custody and timelocked governance, connects via interoperable permissioned ledgers toward a BIS-style unified ledger, and tracks a KPI dashboard where atomic-settlement share exceeds 80%, fails approach zero, and over 90% of value settles in central-bank or fully-reserved money. That is the summit — and the map in these five chapters is how you climb to it.

12 · REFERENCE

Glossary & Further Reading

A fast reference to the vocabulary used across this guide, followed by the sources a serious digital-assets professional is expected to know.

A. Glossary of key terms

Ledgers & tokens

Token — a claim or asset represented as a transferable entry on a programmable ledger, with rules enforced in code.

Distributed ledger technology (DLT) — a shared, synchronised record maintained across multiple parties rather than in one institution's database; a blockchain is one type.

Permissioned vs permissionless — a permissioned ledger restricts who can validate and transact (the institutional norm); a permissionless one is open to anyone.

Smart contract — self-executing code on a ledger that moves tokens and enforces terms automatically (e.g. pays a coupon on a date).

Atomic settlement — two legs of a transaction settle simultaneously and indivisibly, or neither does, eliminating the gap where one party has paid but not received.

Delivery-versus-payment (DvP) / payment-versus-payment (PvP) — settlement mechanisms that link asset delivery to payment (DvP) or two currency payments (PvP) so neither can fail alone.

Oracle — a service that feeds off-chain data (prices, rates, events) to a smart contract; a point of trust and risk.

Custody / private key / MPC — control of a token rests on its cryptographic private key; multi-party computation (MPC) splits the key so no single party holds it.

Finality — the point at which a transfer is irrevocable, in both the technical and the legal sense.

Digital money

Stablecoin — a privately issued token designed to hold a stable value, usually by being backed by reserves (fiat-backed), over-collateralised with crypto, or — now largely discredited — algorithmically.

Peg / depeg — the target price (usually 1:1 with a fiat unit) and the event of trading away from it; held by mint/redeem arbitrage.

Tokenized deposit — a commercial-bank deposit represented on a ledger, redeemable at par 1:1 and staying inside the regulated banking system.

Central bank digital currency (CBDC) — a direct central-bank liability in digital form; retail (for the public) or wholesale (for interbank/cross-border settlement).

Singleness of money — the principle that all forms of money trade at par with one another, so a dollar is a dollar whatever its form.

Two-tier model — the design in which the central bank issues the money and regulated intermediaries distribute and service it.

Tokenized assets, markets & rules

Real-world asset (RWA) tokenization — representing off-chain assets (bonds, funds, credit, real estate) as tokens on a ledger.

Fractionalization — dividing an asset into small token units, lowering minimum investment and widening access.

Tokenized money-market fund — a regulated money fund whose shares are issued and transferred on-chain, used as yield-bearing on-chain cash and collateral.

Collateral mobility — the speed and ease with which collateral can be moved and re-used; tokenization cuts it from a day to minutes.

Intraday repo — secured borrowing for hours rather than overnight, made practical by instant on-chain settlement.

Unified ledger — the BIS vision of central-bank money, commercial-bank money and tokenized assets on a common programmable platform.

Travel rule — the FATF requirement that originator/beneficiary information travel with a transfer, for AML/KYC.

Basel crypto-asset standard — the prudential rules splitting exposures into Group 1 (tokenized traditional assets / qualifying stablecoins, roughly like the underlying) and Group 2 (e.g. unbacked crypto, up to a 1250% risk weight — near dollar-for-dollar capital).

MiCA / GENIUS Act — the EU's Markets in Crypto-Assets regulation and the US federal stablecoin law, setting reserve, redemption and disclosure rules.

B. Further reading — the canon

- Bank for International Settlements — the *Annual Economic Report* chapters on tokenization and the “unified ledger”, and BIS Innovation Hub projects (Agorá, mBridge, Pine, Helvetia).
- Hyun Song Shin and BIS research — the conceptual framework for tokenization, the money flower and the singleness of money.
- Financial Stability Board & IOSCO — reports on stablecoins, crypto-asset activities and tokenization.
- Basel Committee — the prudential treatment of banks' crypto-asset exposures.
- MiCA (EU) and the GENIUS Act (US) — the primary stablecoin and crypto-asset rulebooks.

- Platform and issuer documentation — JPMorgan Kinexys, BlackRock BUIDL / Securitize, Franklin Templeton, Broadridge DLR, HQLAx, Fnality, SIX SDX, DTCC.
- Project reports on landmark issues — World Bank bond-i, the EIB digital bonds, and the Hong Kong tokenized green bonds.

DISCLAIMER

This guide is an educational reference, not investment, legal or tax advice, and takes no view on any token or crypto-asset as an investment. Figures in worked examples are illustrative and chosen to teach method; market sizes and project details move quickly and are stated as approximate. Rules (MiCA, the GENIUS Act, the Basel standard) vary by jurisdiction and evolve; always check the current text before relying on it.